

Siril Dukkupati, PhD

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Education

McGill University, Montréal, Canada

Jan 2022 – Aug 2025

PhD in Mechanical Engineering (Biomechanics and MedTech focus)

- Secured CAD 100K Fonds de recherche du Québec Doctoral Scholarship, CAD 111K McGill Engineering Doctoral Award, and Graduate Excellence Awards for 5 consecutive years
- Published 4 peer-reviewed journal articles, 1 preprint, and presented at 25+ international conferences

MS in Mechanical Engineering

Sept 2020 – Dec 2021

- Awarded CAD 7K Fonds de recherche du Québec Master's Scholarship and 2 international travel grants
- GPA: 4.0/4.0; fast-tracked into PhD

Manipal University, Manipal, India

Sept 2016 – May 2020

BTech in Mechanical Engineering (Minor in Design)

- Manipal Scholar Award (top 1% of department, CAD 25K) and two competitive summer research fellowships in Canada and Belgium (CAD 13K total)
- GPA: 9.17/10.0, graduated with distinction

Experience

Course Lecturer

Montréal, CA

McGill University

Jan 2024 – Jan 2025

- Designed and delivered MECH 561: Biomechanics of Musculoskeletal Systems to ~50 graduate and undergraduate students across 2 semesters.
- Developed lesson plans, supervised group projects, and created assessments (midterm and final exam), ensuring alignment with program learning goals. Excellent feedback from students in course evals (80%+).

Biomedical engineer

Montréal, CA

Simcare Tech.

Aug 2024 – Aug 2025

- Simcare is a lab spin-off based on the technology I developed during my doctoral work.
- Developed custom spine models for industry (1) and academic (1) partners for surgical training and testing.

CTO, Mechanical Engineer

Bangalore, IN

Foodlabs Inc.

Jul 2020 – Sept 2024

- Directed a 4-member engineering team to design and deploy an AI-enabled robotic kitchen ecosystem, launching 3 products (2 hardware, 1 software) in public beta.
- Worked on 2 patents (pending) with the Indian Patent Office, strengthening company's IP portfolio.

Research Collaborations

MDsim Inc.

Jul 2024 - present

Title: Experimental Validation of SpineSim Digital Twin Platform Using Analogue Spine Models

- Generated validation data for the “SpineSim” digital twin platform using experimental testing of patient-specific spine phantoms, enabling adoption in surgical planning.
- Posteriolateral Lumbar Fusion (PLF) surgical simulation using **Medtronic** CD HORIZON Spinal System
- Coordinated with engineers and surgeons to deliver all milestones within a 3-month timeline. 1 journal article currently under preparation.

International Spine Biomechanics Consortium

Apr 2022 - present

Title: Towards Standardization of Experimental Testing Practices in Spine Biomechanics

- Formed a working group of 7 top international spine biomechanics research labs (16 researchers) focused on standardizing experimental methods in spine biomechanics.
- Presented findings at top biomechanics conferences like ORS and ISTA. 2 journal articles under preparation.

Wake Forest University

Oct 2023 - present

Title: New Techniques in Multi-dimensional Spine Biomechanics Characterization

- Partnered with Dr. Philip Brown's lab to conduct multi-lab, multi-specimen robotic spine testing using a KUKA KR300 robotic arm.
- Secured funding from the Orthopaedic Research Society in 2024. Presented at 2 conferences and 1 journal article currently under preparation.

Montreal General Hospital

July 2025 - present

Title: Performance of Percutaneous Pedicle Screw Fixation System

- Conducted mechanical testing using the **Johnson & Johnson** VIPER PRIME percutaneous pedicle screw system to evaluate how insertion depth affects pullout strength and insertion torque.
- Led project planning, timeline management, and resource coordination to complete all experimental milestones within a 6-week window.
- Supervised test execution, data analysis, and reporting, ensuring technical accuracy while facilitating collaboration between multiple clinical advisors and spine surgeons.

Publications

4 journal articles | 2 open-source models | 25 international conferences | 4 manuscripts in preparation

Complete list available on [ResearchGate](#)

Leadership

Vice President of Academics, GAMES (2022–2023) – Represented 150+ graduate students, secured permanent stipend increases, and strengthened engagement between faculty and students

TA Union Representative, AGSEM (2023–2024) – Elected to advocate for 50+ teaching assistants, negotiating fair workload and contract practices with university administration.

Judge, SURE Program (2022–2025) – Evaluated 15+ undergraduate research projects annually, mentoring students on research projects.

3D Printing Supervisor, MBR Lab (2020–2025) – Trained 10+ researchers and established lab-wide protocols for fabrication and safety compliance.

Research Team Lead, Mars Rover Manipal (2016–2019) – Led 25+ member robotics team to win **1st place** at Indian Rover Challenge 2017 and **7th globally** at University Rover Challenge 2018.

Skills and Interests

Technical skills: MATLAB, Python, LabVIEW, R, 3D Printing; Project management tools (Trello, Confluence); experienced in leading cross-disciplinary teams.

Interests: Cooking, intramural badminton and cricket, photography, creative web dev projects.